

# Introduction

Humans, together with animals and other living things, form what is called the biosphere—the living part of the world. Since modern humans first appeared in Africa about 200,000 years ago, we have colonized virtually the entire world—even hot deserts and the ice-cold Arctic. As we have done so, our impact on the biosphere has been far-reaching.

## Human impact

The human “footprint” on planet Earth is deep and broad. We have transformed the landscape—clearing forests to produce food, digging minerals and ores from the ground, and channeling and storing water to meet our needs. Our living space is concentrated into larger and larger cities, but these cities are hungry for food and energy taken from the surrounding land.



### Renewable energy

New ways of harnessing the energy of sunlight and wind are reducing our use of fossil fuels. Unlike fossil fuels, these energy sources will never run out.

### Natural resources

Buried within Earth’s crust there are limited supplies of minerals, metal ores, and fossil fuels (coal, oil, and gas). Once these reserves are exhausted, they cannot be replaced. Burning these fuels also damages Earth’s atmosphere and is contributing to global warming.



## Population

For most of humanity’s existence, the human population grew relatively slowly. In 10,000 BCE, there were only 1–5 million people on Earth. By 1000 BCE, after farming was invented, the population had increased to about 50 million. Since reaching the 1 billion mark in 1804, during the early Industrial Revolution, the population has expanded much more quickly than ever before.

### Growing bigger, fast

The period since the late 1950s has seen the human population more than double.

